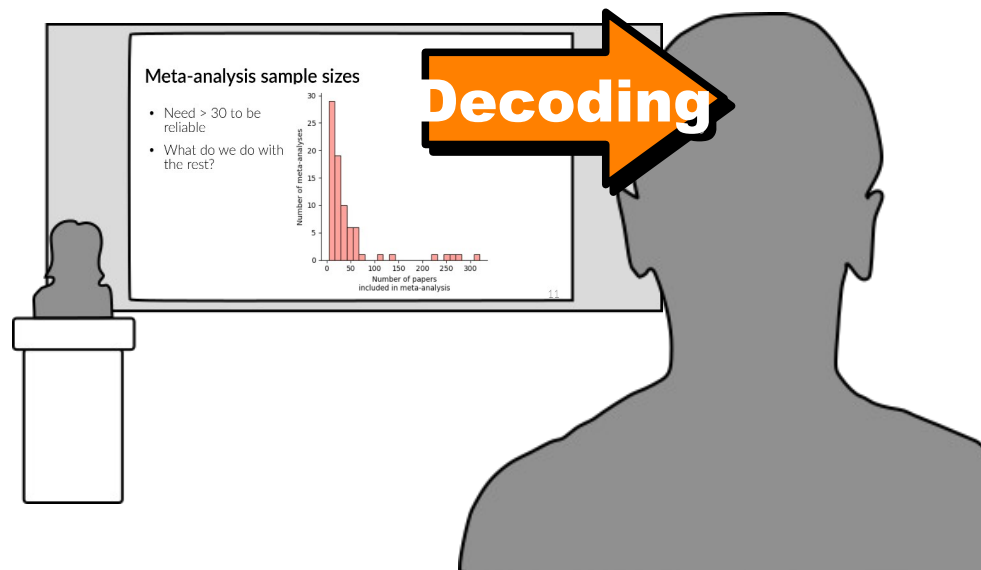


Introduction to Data Visualization

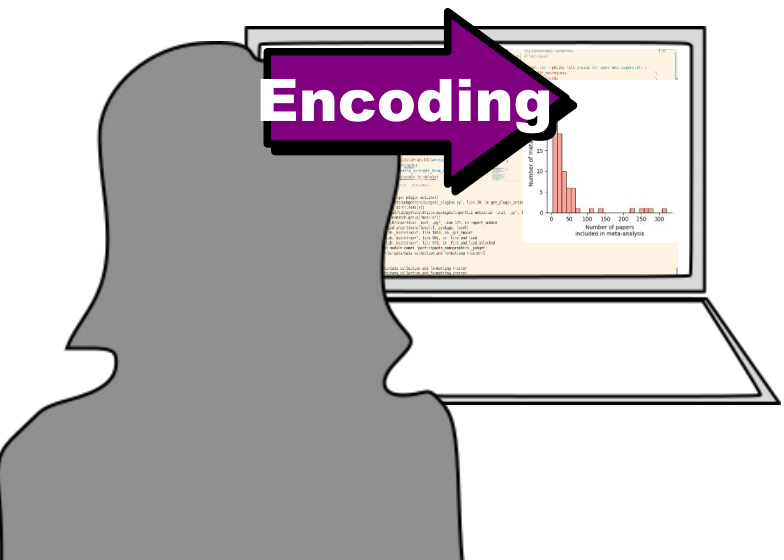
Part 1: Decoding

Johanna Bayer
Slides by Kendra Oudyk

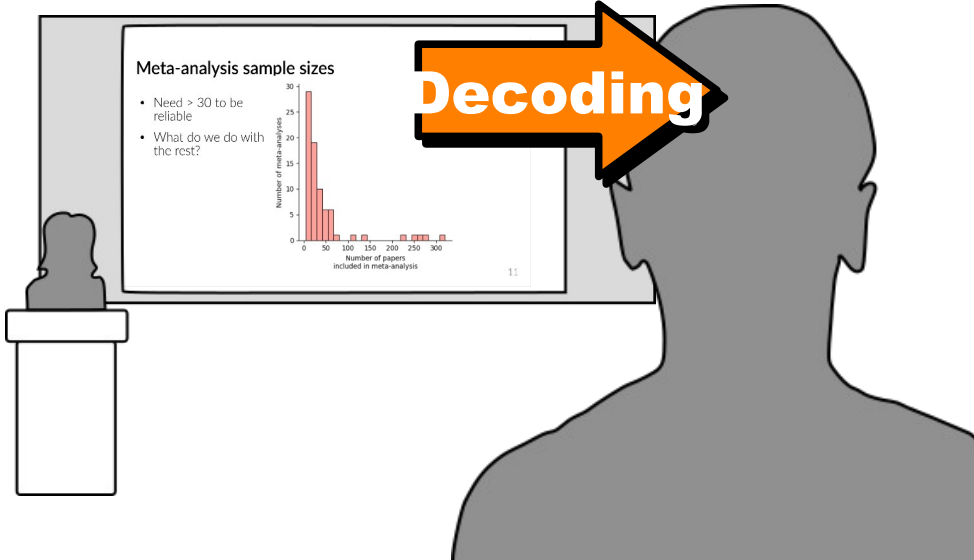


Goal

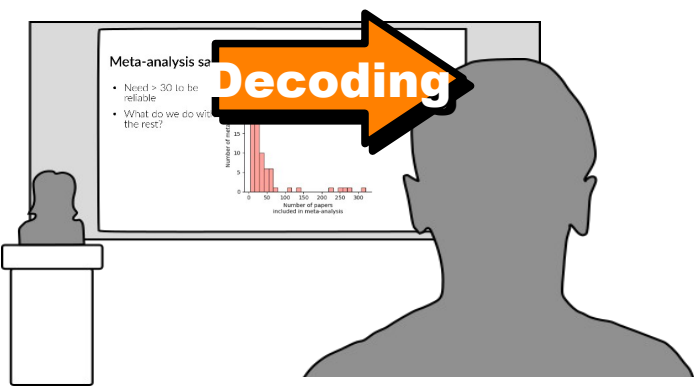
Use principles of visual **encoding** and **decoding** to **efficiently** create visualizations that are **effective** and **reproducible**



Encoding



Decoding



To plan an **effective visualization**, we need to think about

- **Message**

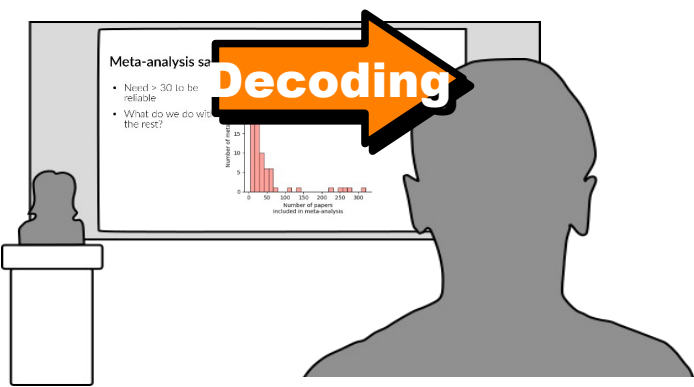
- **What** we want to communicate

- **Perception**

- **How best** to communicate it

- **Conventions**

- **How** it's **usually** communicated



To plan an **effective visualization**, we need to think about

- **Message**

- **What** we want to communicate

- **Perception**

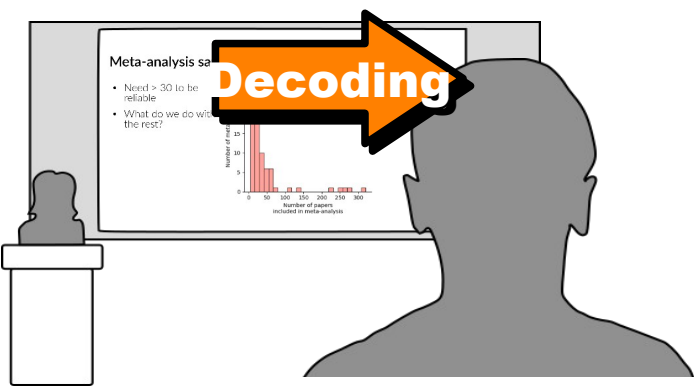
- **How best** to communicate it

- **Conventions**

- **How** it's **usually** communicated

Message

- Raw data has no message
- Abstract it
 - “There are more males than females in science”
--> A difference between magnitudes
 - “These brain areas activate together”
--> A grouping / pattern



To plan an **effective visualization**, we need to think about

- **Message**

- **What** we want to communicate

- **Perception**

- **How best** to communicate it

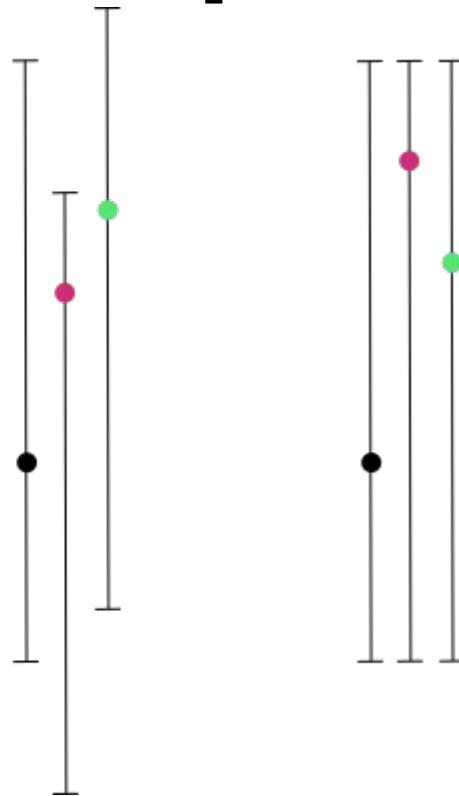
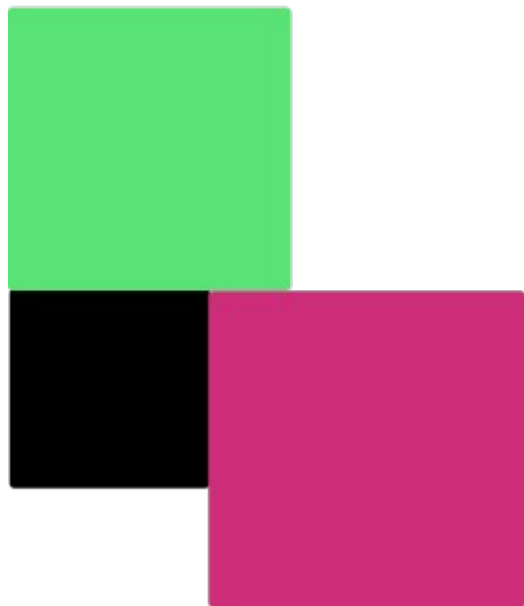
- **Conventions**

- **How** it's **usually** communicated

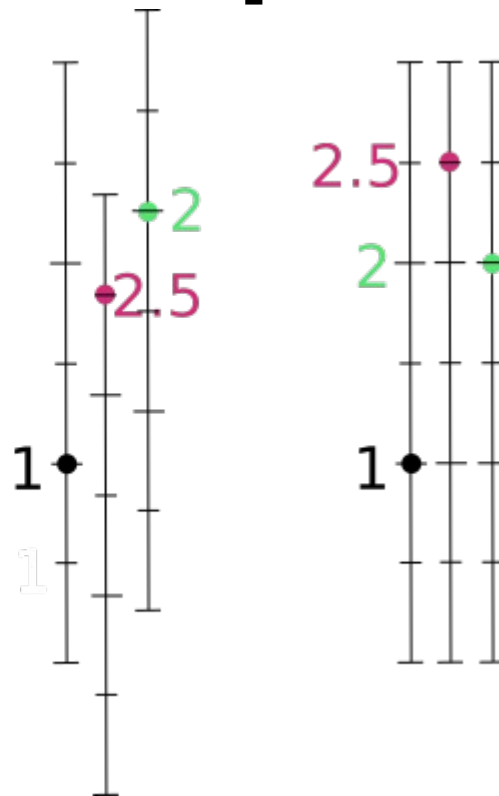
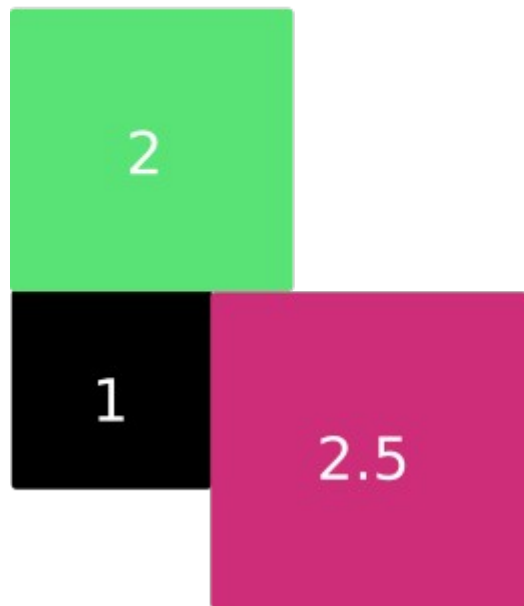
Which blue bar is bigger?



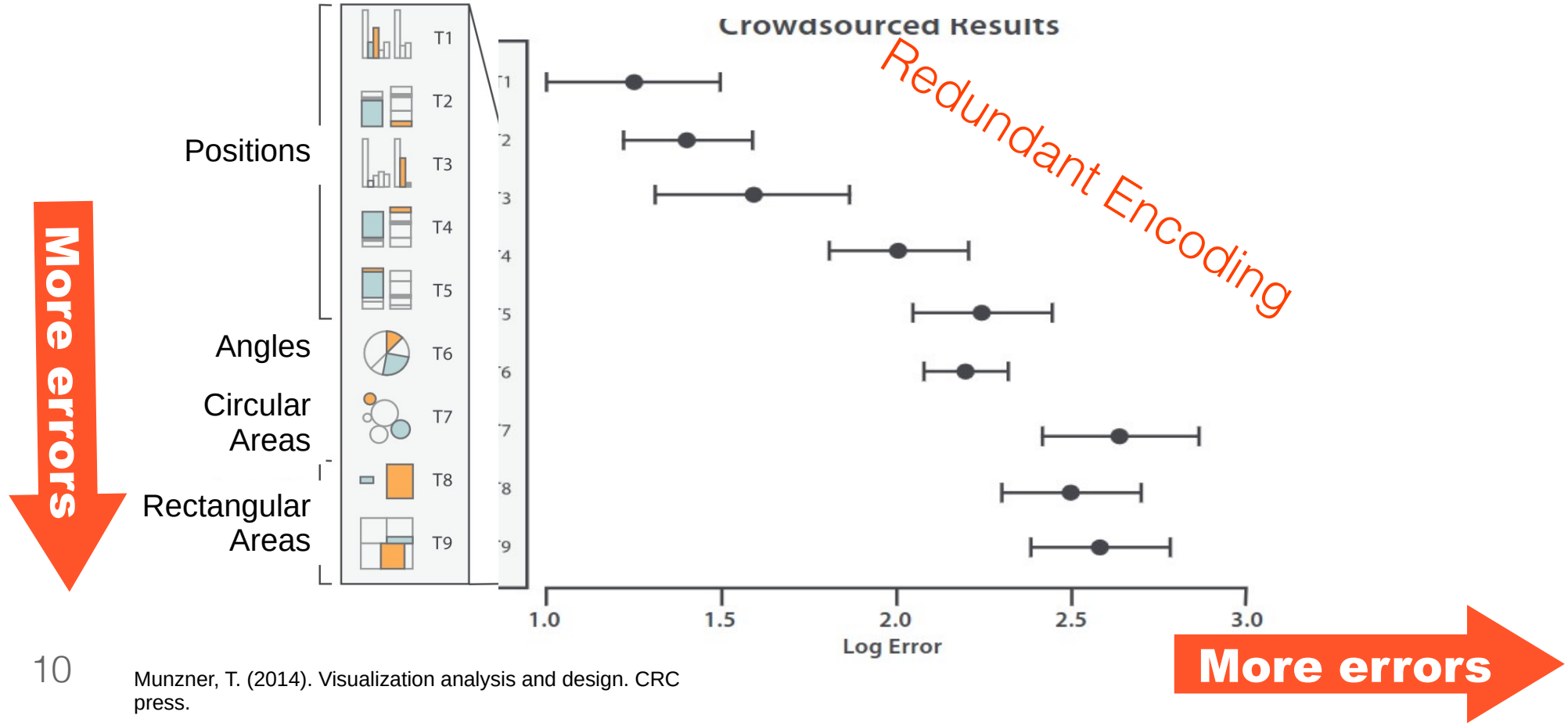
Which is 2x black, green or pink?



Which is 2x black, green or pink?



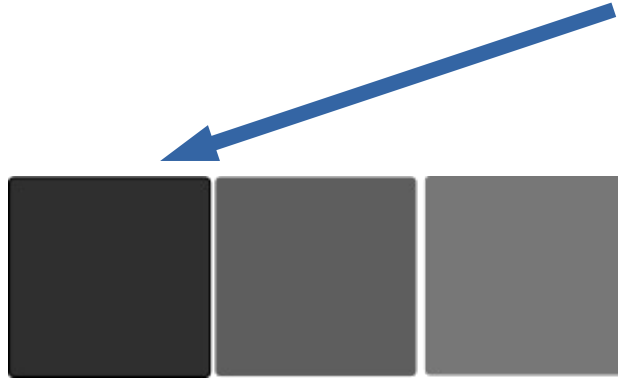
We're better at judging aligned positions



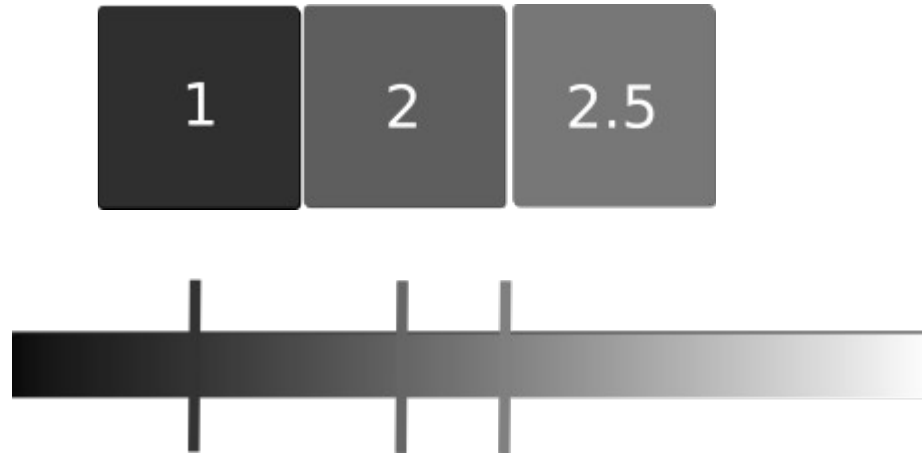
Decoding values vs. patterns

- “There are more males than females in science”
--> A difference between **magnitudes**
- “These brain areas activate together”
--> A grouping / **pattern**

Which is 2x lighter than the leftmost box?

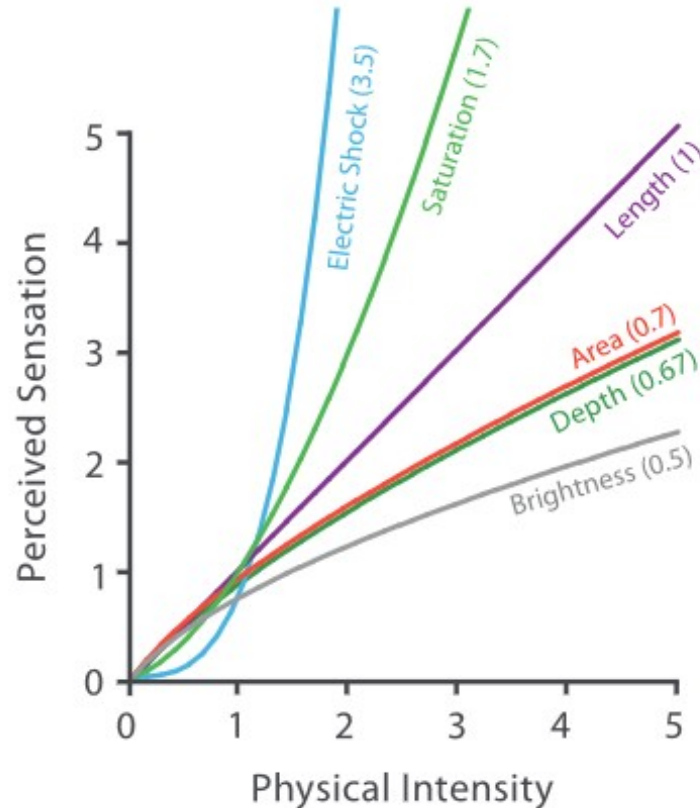


Which is 2x lighter than the leftmost box?



Physical intensity vs perceived intensity

Steven's Psychophysical Power Law: $S = I^N$



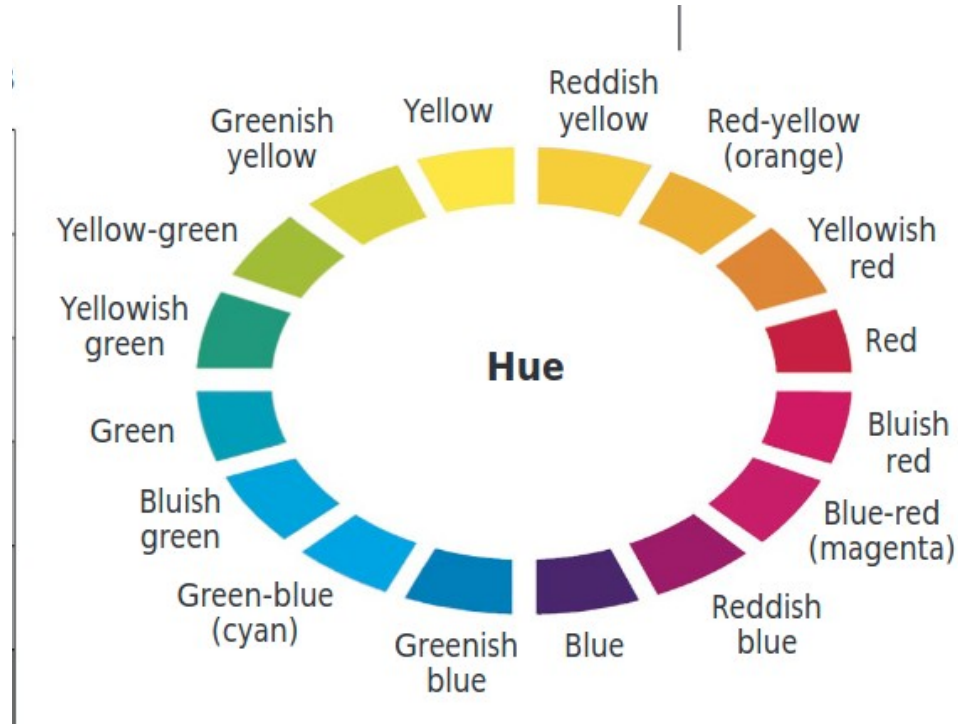
Color: salient but complicated



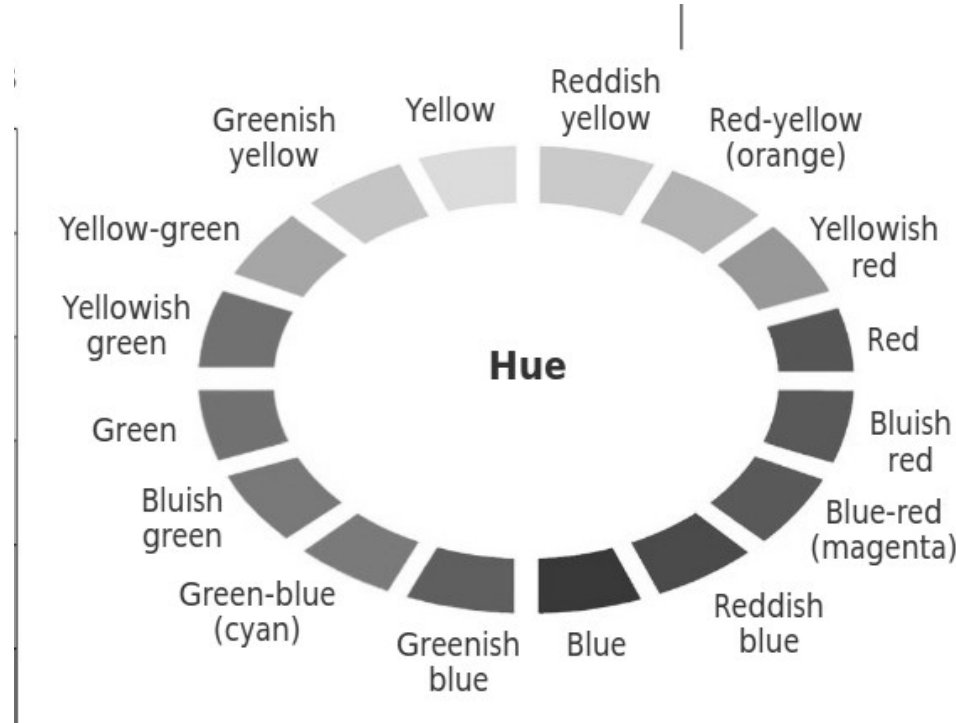
The HSL color model

- H - Hue: the "type" of color, represented as an angle from 0–360°. Red at 0°, green at 120°, blue at 240°, and back to red at 360°
- S - Saturation: 0% is grey, 100% is the fully vivid version of that hue
- L – Lightness/Luminance: 0% is black, 50% is the "pure" color, 100% is white

Hue: how we talk about color

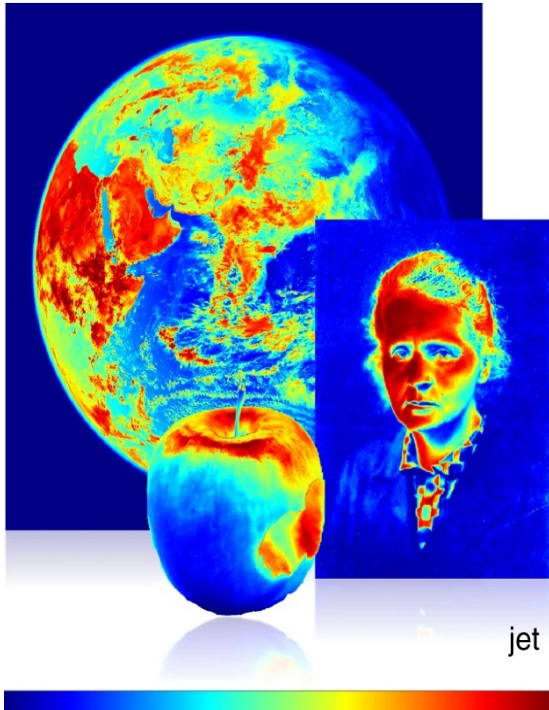


Luminance: an important subconscious cue



Named colors don't work well for ordered values

b

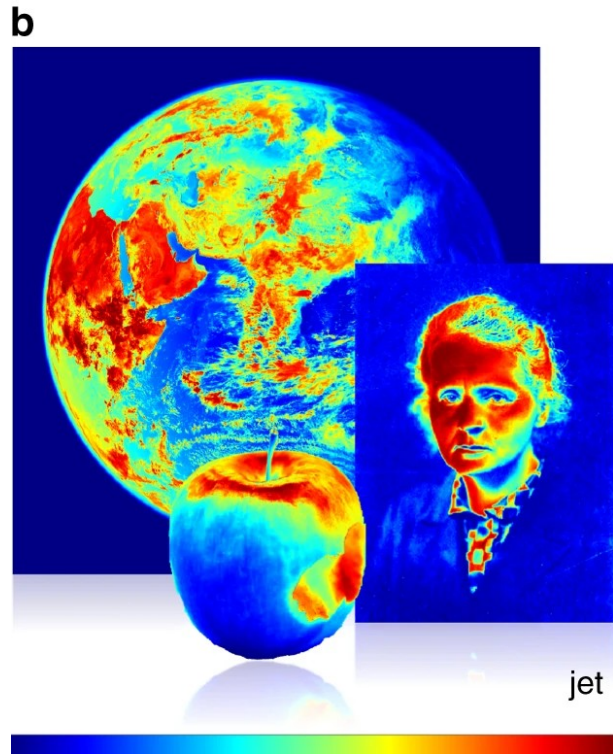


Named colors don't work well for ordered values

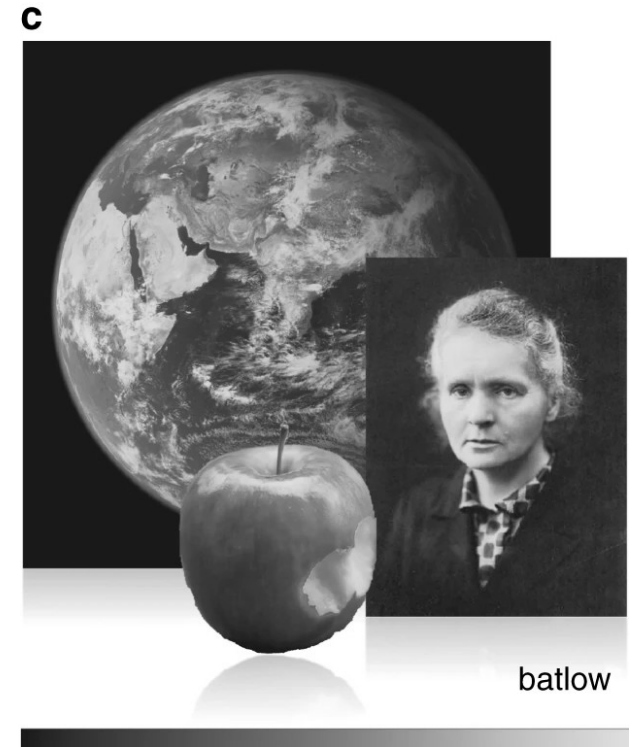
b



Certain colormaps do work



Certain colormaps do work



The brain's sensitivity to luminance

Which is
lightest?



Physical
Lightness



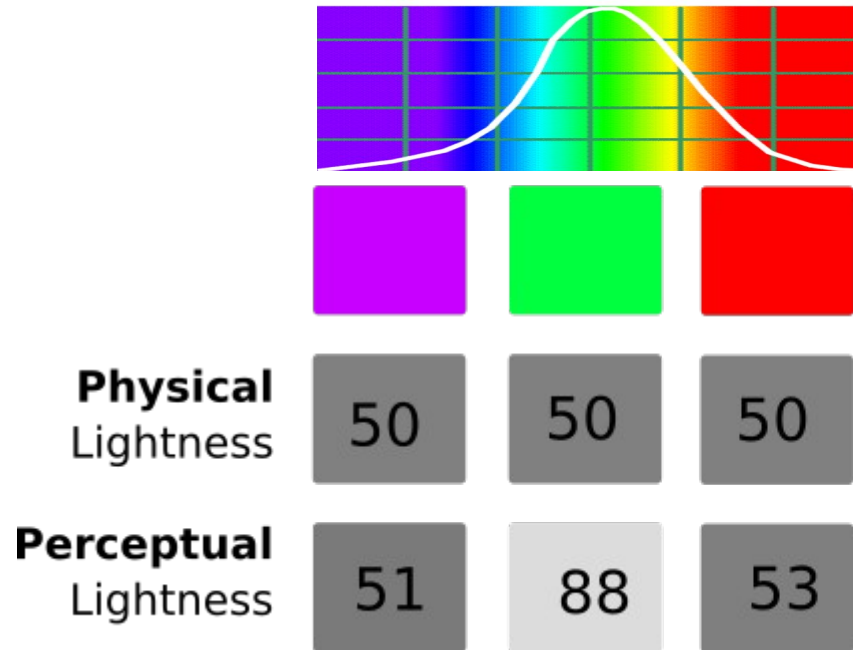
50

50

50

			
Physical Lightness			
Perceptual Lightness			

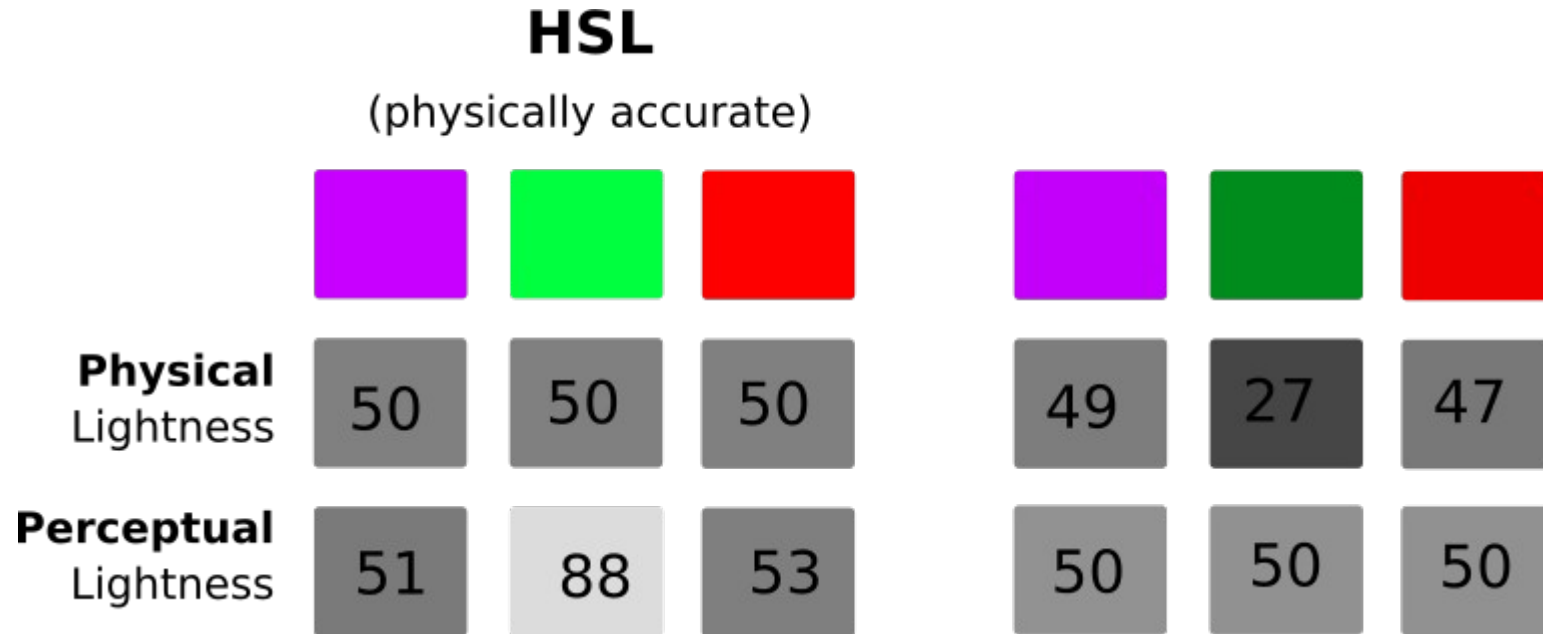
Spectral sensitivity to luminance (lightness)



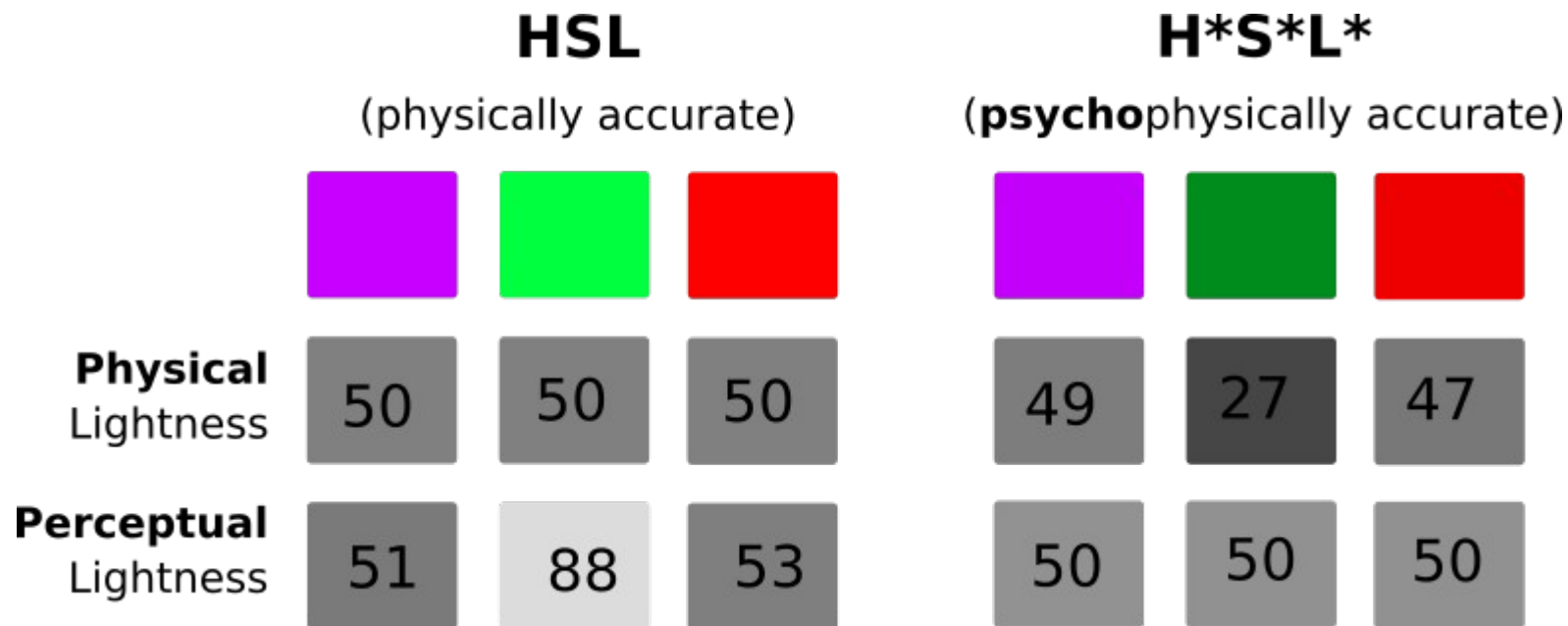
Spectral sensitivity to luminance (lightness)

	HSL (physically accurate)		
			
Physical Lightness	 50	 50	 50
Perceptual Lightness	 51	 88	 53

Spectral sensitivity to luminance (lightness)



Spectral sensitivity to luminance (lightness)



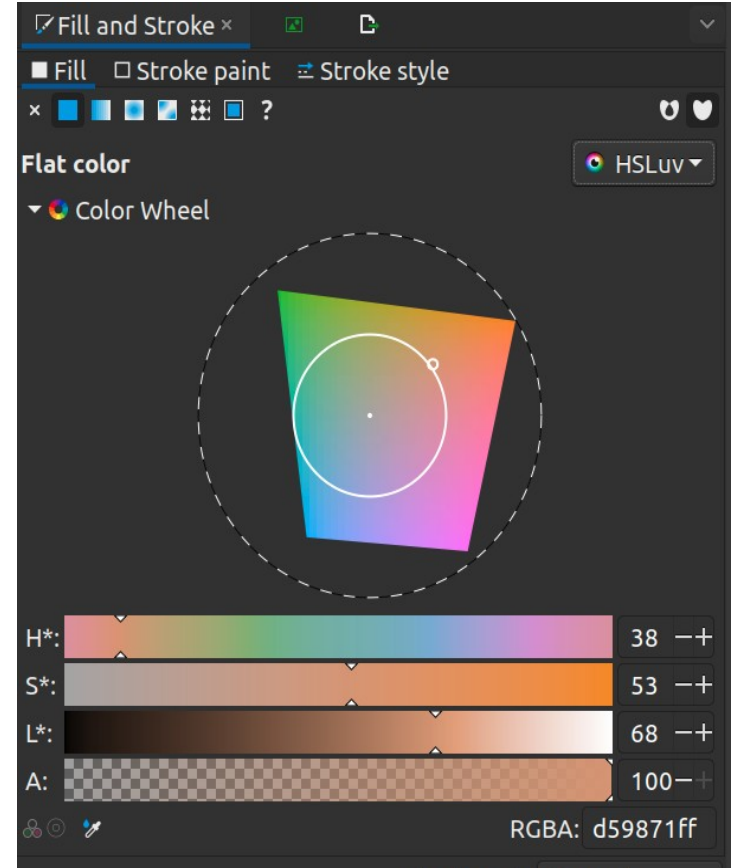
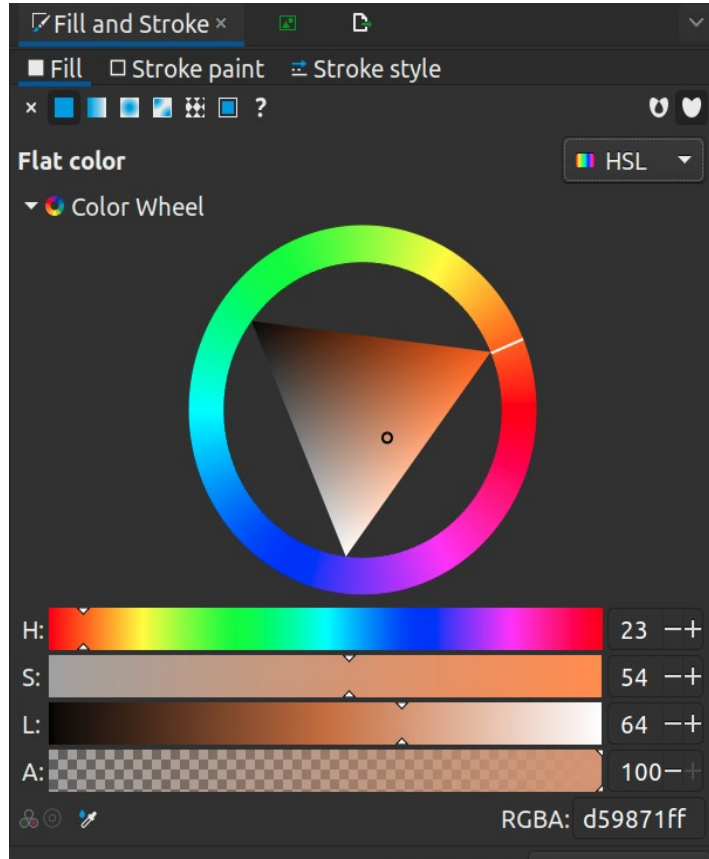
batlow



HSL

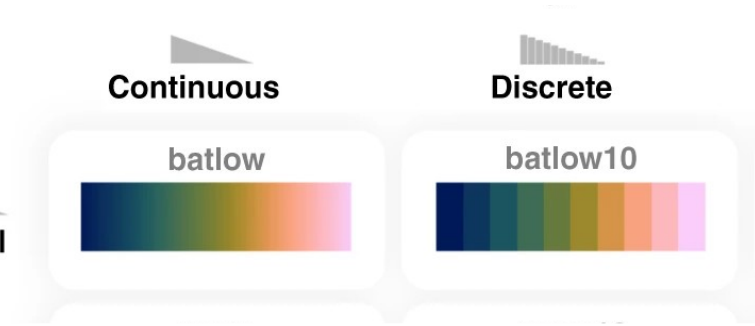
vs H*S*L* (or HSLuv)

Hue
Saturation
Lightness / Luminance



Types of colormaps

Sequential



Types of colormaps

Sequential

Diverging

Continuous

Discrete

batlow



batlow10



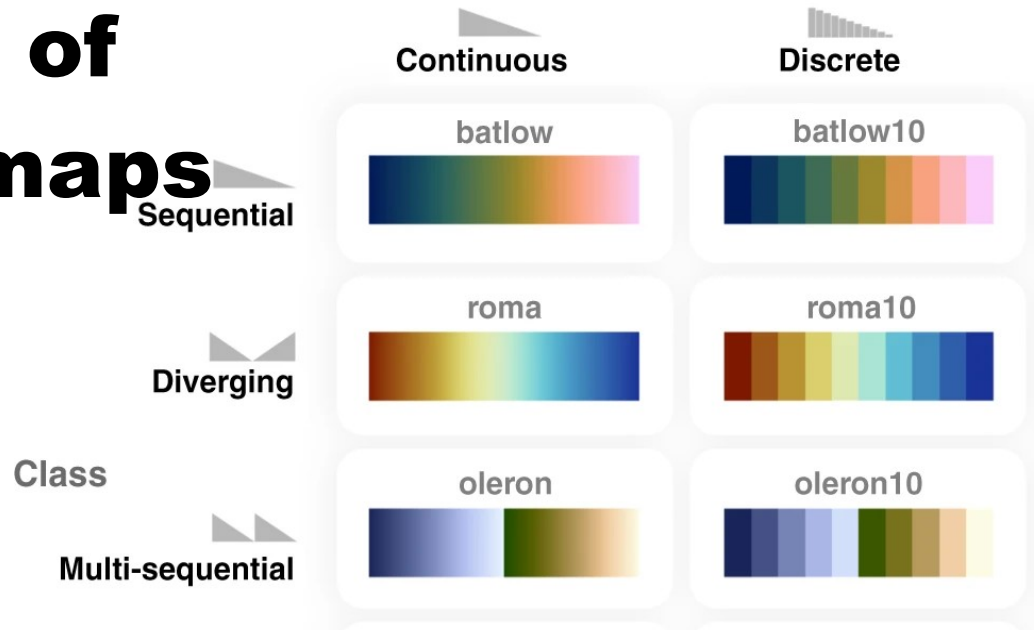
roma



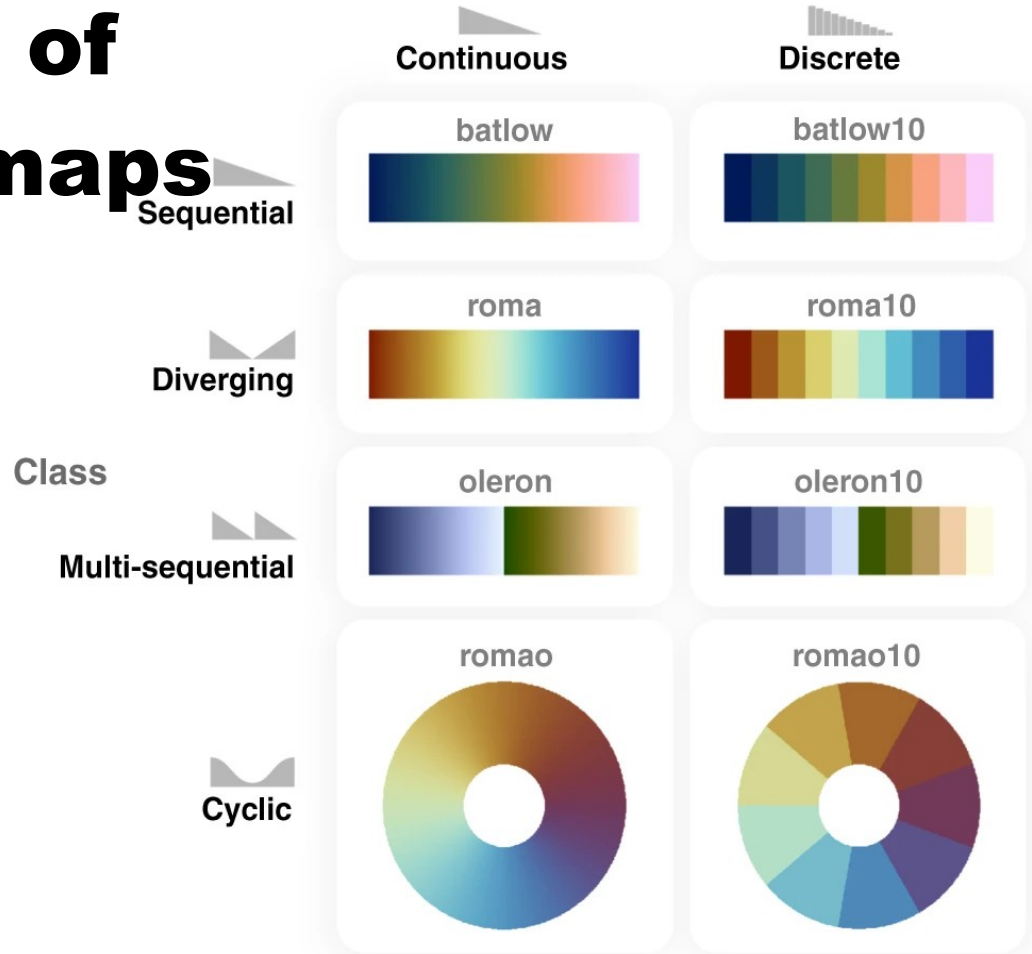
roma10



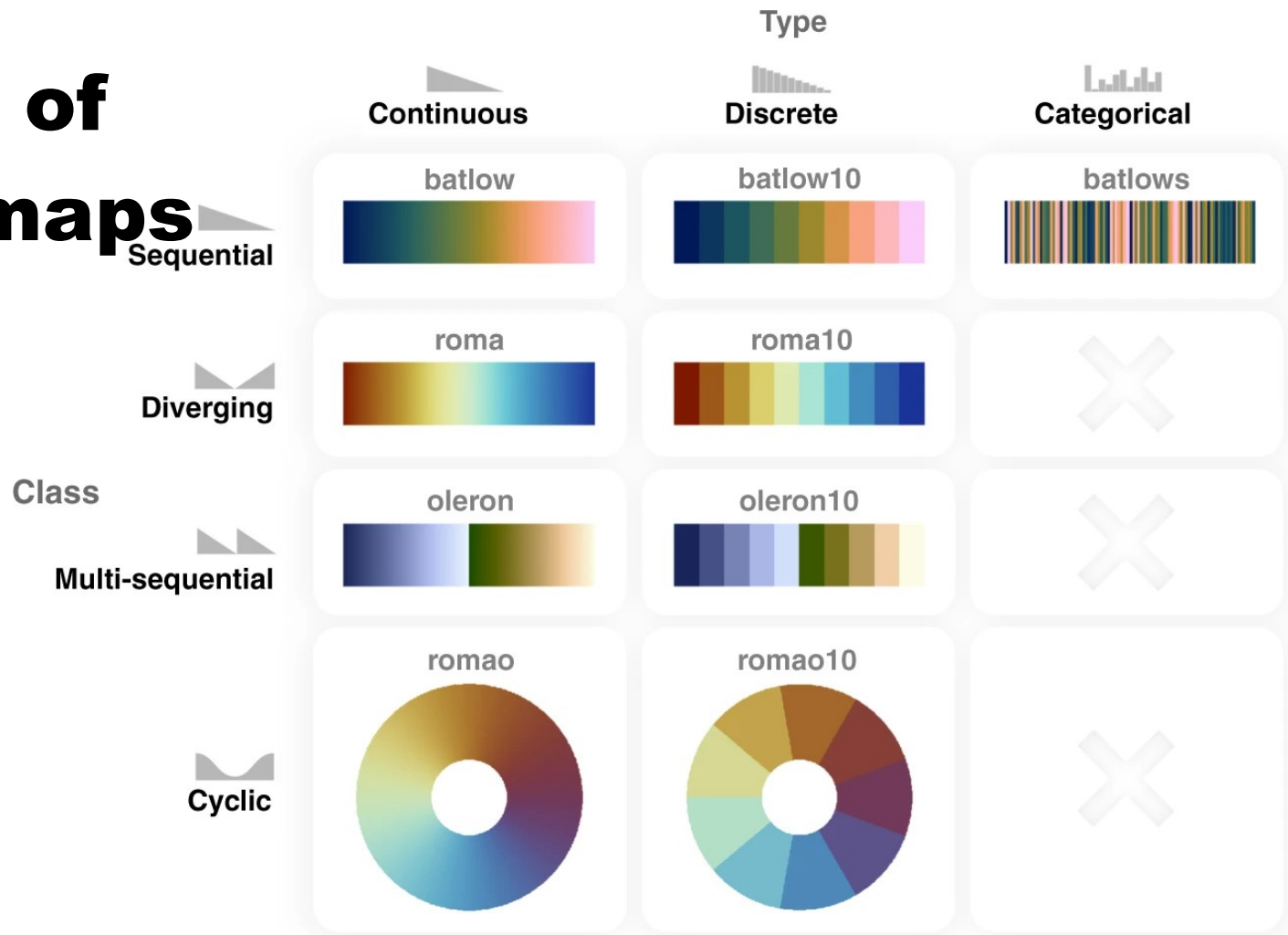
Types of colormaps



Types of colormaps

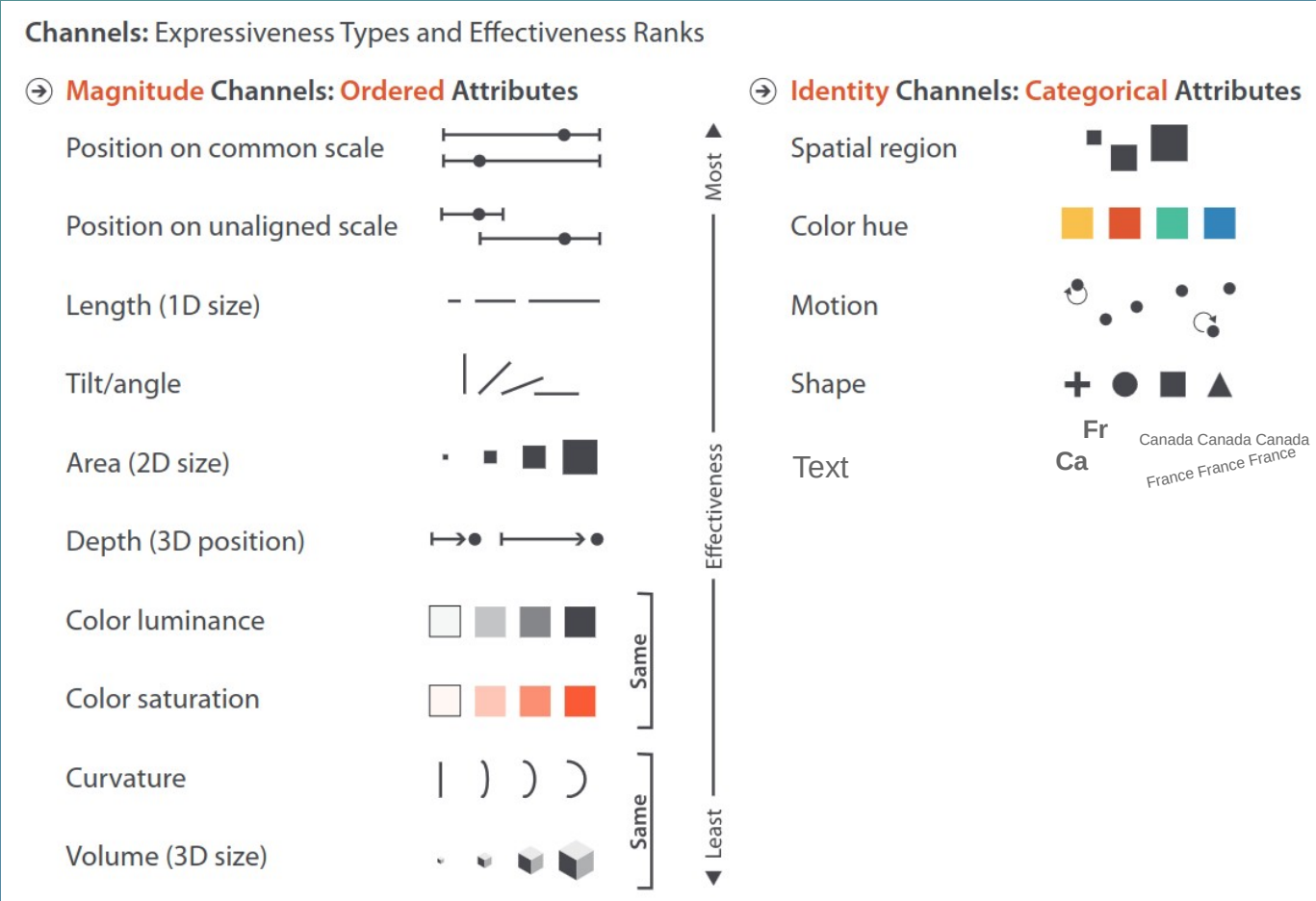


Types of colormaps



(For future reference)

Choosing effective charts



https://matplotlib.org/cheatsheets/_images/cheatsheets-1.png

Munzner, T. (2014). Visualization analysis and design. CRC press.

Quick start

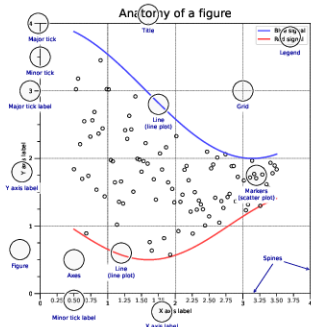
```
import numpy as np
import matplotlib as mpl
import matplotlib.pyplot as plt
```

```
X = np.linspace(0, 2*np.pi, 100)
Y = np.cos(X)
```

```
fig, ax = plt.subplots()
ax.plot(X, Y, color='green')
```

```
fig.savefig("Figure.pdf")
fig.show()
```

Anatomy of a figure



Subplots layout

```
subplot[s](rows,cols,...)
fig, axs = plt.subplots(3, 3)
```

```
G = gridspec(rows,cols,...)
ax = G[0,:]
```

```
ax.inset_axes(extent)
```

```
d=make_axes_locatable(ax)
ax = d.new_horizontal('10%')
```

Getting help

- matplotlib.org
- github.com/matplotlib/matplotlib/issues
- discourse.matplotlib.org
- stackoverflow.com/questions/tagged/matplotlib
- https://git.io/matplotlib/matplotlib
- twitter.com/matplotlib
- Matplotlib users mailing list

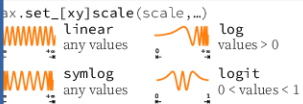
Basic plots



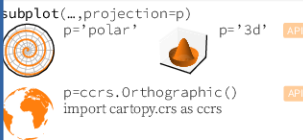
Advanced plots



Scales



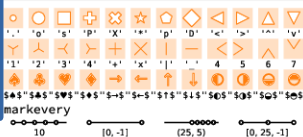
Projections



Lines



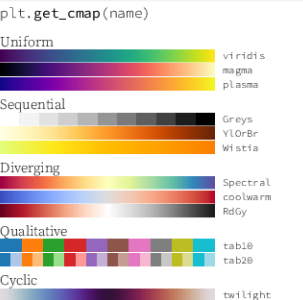
Markers



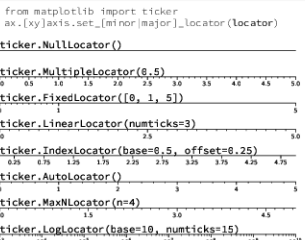
Colors



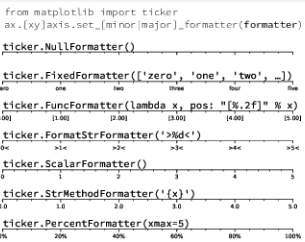
Colormaps



Tick locators

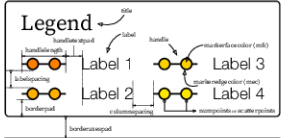


Tick formatters



Ornaments

```
ax.legend(...)
```



```
ax.colorbar(...)
```



```
ax.annotate(...)
```



Event handling

```
fig, ax = plt.subplots()
def on_click(event):
    print(event)
fig.canvas.mpl_connect('button_press_event', on_click)
```

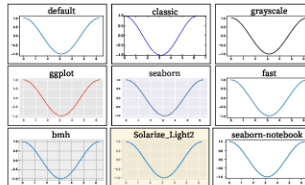
Animation

```
import matplotlib.animation as mpla

T = np.linspace(0, 2*np.pi, 100)
S = np.sin(T)
line, = plt.plot(T, S)
def animate(i):
    line.set_ydata(np.sin(T+i/50))
anim = mpla.FuncAnimation(
    plt.gcf(), animate, interval=5)
plt.show()
```

Styles

```
plt.style.use(style)
```



Quick reminder

```
ax.grid()
ax.set_xlim(vmin, vmax)
ax.set_ylabel(label)
ax.set_yticks(ticks, [labels])
ax.set_yticklabels(labels)
ax.set_title(title)
ax.tick_params(width=10, ...)
ax.set_axis_on/off()
```

```
fig.suptitle(title)
fig.tight_layout()
plt.gcf(), plt.gca()
mpl.rc('axes', linewidth=1, ...)
[fig|ax].patch.set_alpha(0)
text=r'$\frac{e^{\pi i p}}{2^n}$'
```

Keyboard shortcuts

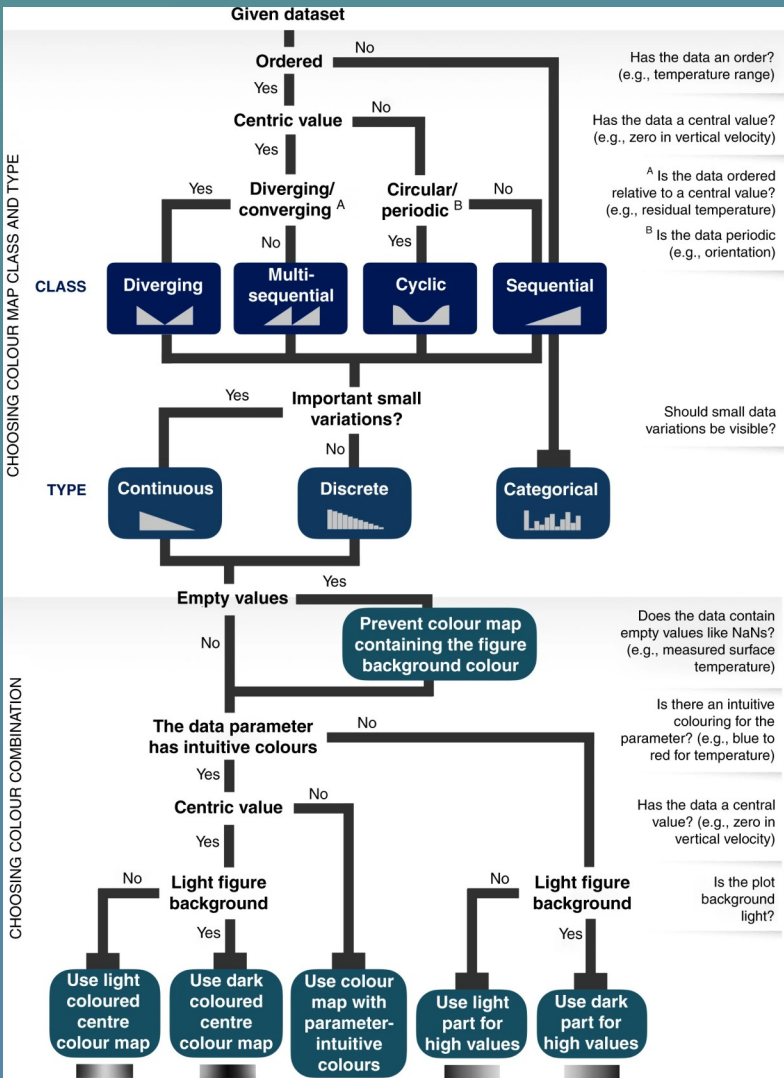
Ctrl+S	Save	Ctrl+W	Close plot
r	Reset view	f	Fullscreen 0/1
f	View forward	b	View back
p	Pan view	o	Zoom to rect
x	X pan/zoom	y	Y pan/zoom
g	Minor grid 0/1	G	Major grid 0/1
l	X axis log/linear	L	Y axis log/linear

Ten simple rules

1. Know your audience
2. Identify your message
3. Adapt the figure
4. Captions are not optional
5. Do not trust the defaults
6. Use color effectively
7. Do not mislead the reader
8. Avoid "chartjunk"
9. Message trumps beauty
10. Get the right tool

(For future reference)

Choosing effective colormaps



Activities

Firefox Web Browser

Firefox

Calendar

Terminal

Globe

Info

Frontiers

rogowitz a

IEEE Xplor

Data visual

IEEE Xplor

Fig. 2: Colo

Choosin x

https://matplotlib.org/stable/tutorials/colors/colormaps.html#sphx-gl-r-tutorials-colors-colorma

stable

matplotlib

Plot types Examples Tutorials Reference User guide Develop Releases

Section Navigation

Introductory

Intermediate

Advanced

Colors

Specifying colors

Customized Colorbars Tutorial

Creating Colormaps in Matplotlib

Colormap Normalization

Choosing Colormaps in Matplotlib

Text

Toolkits

Provisional

Sequential

For the Sequential plots, the lightness value increases monotonically through the colormaps. This is good. Some of the L^* values in the colormaps span from 0 to 100 (binary and the other grayscale), and others start around $L^* = 20$. Those that have a smaller range of L^* will accordingly have a smaller perceptual range. Note also that the L^* function varies amongst the colormaps: some are approximately linear in L^* and others are more curved.

```
plot_color_gradients('Perceptually Uniform Sequential',  
                    ['viridis', 'plasma', 'inferno', 'magma', 'cividis'])
```

Perceptually Uniform Sequential colormaps

viridis

plasma

inferno

magma

cividis

```
plot_color_gradients('Sequential',  
                    ['Greys', 'Purples', 'Blues', 'Greens', 'Oranges', 'Re  
                    'YlOrRd', 'OrRd', 'PuRd', 'RdPu', 'BuPu',
```

On this page

Overview

Classes of colormaps

Sequential

Sequential2

Diverging

Cyclic

Qualitative

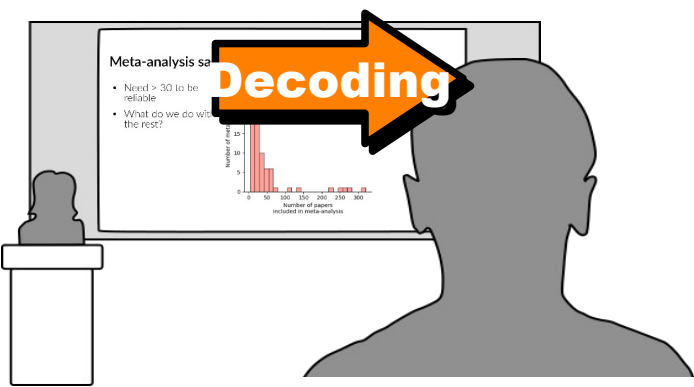
Miscellaneous

Lightness of Matplotlib colormaps

Grayscale conversion

Color vision deficiencies

References



To plan an **effective visualization**, we need to think about

- **Message**

- **What** we want to communicate

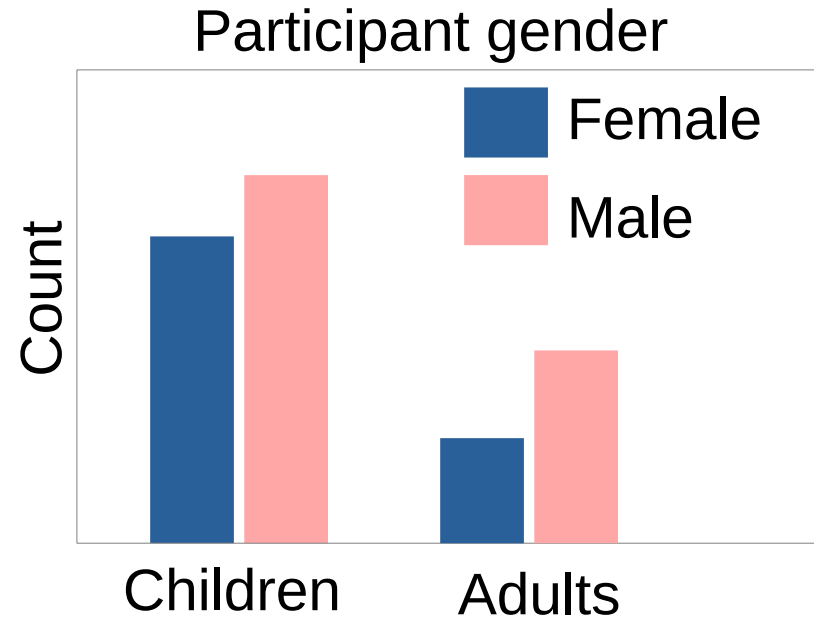
- **Perception**

- **How best** to communicate it

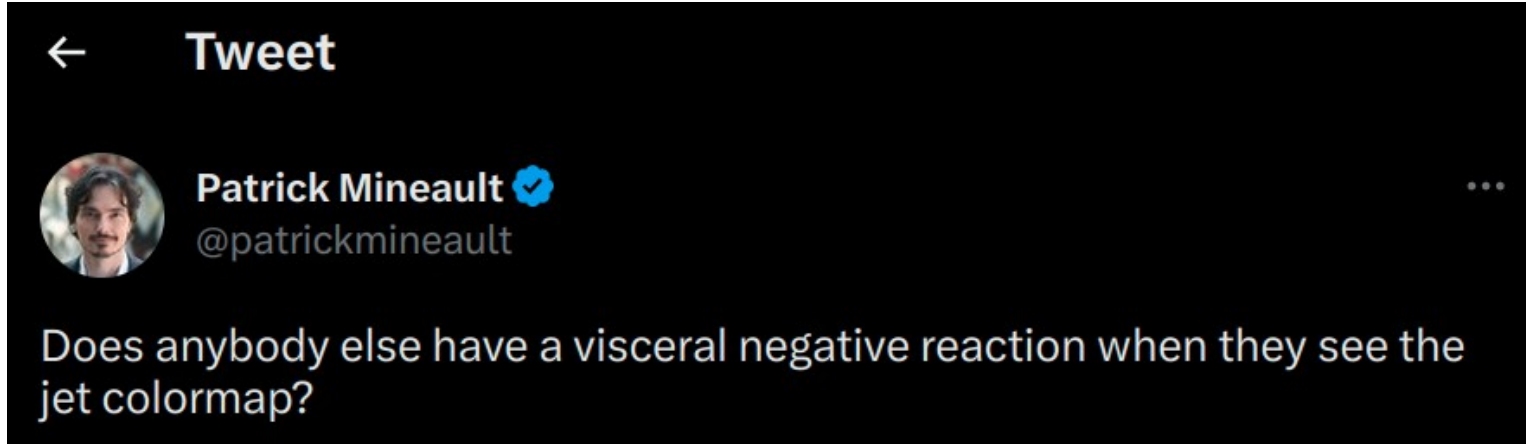
- **Conventions**

- **How** it's **usually** communicated

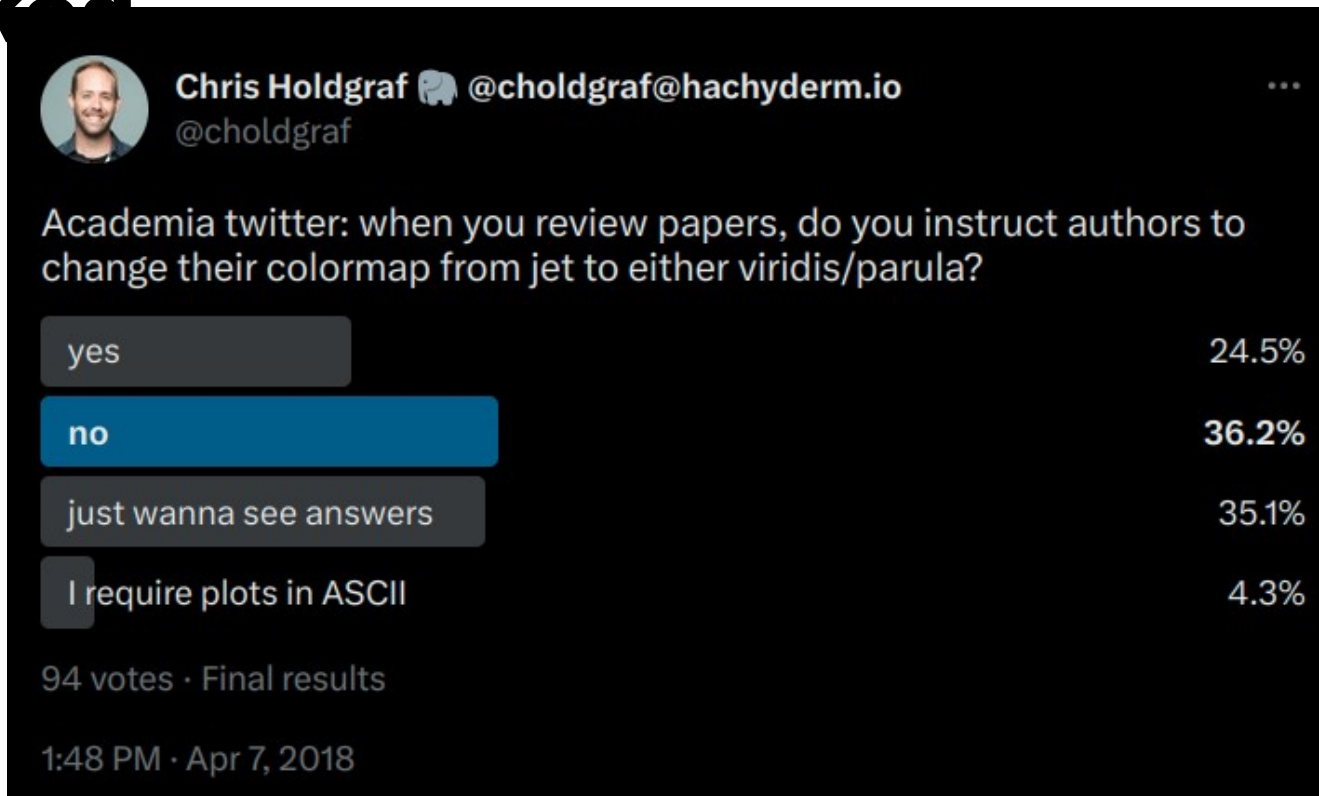
Culture: Color associations



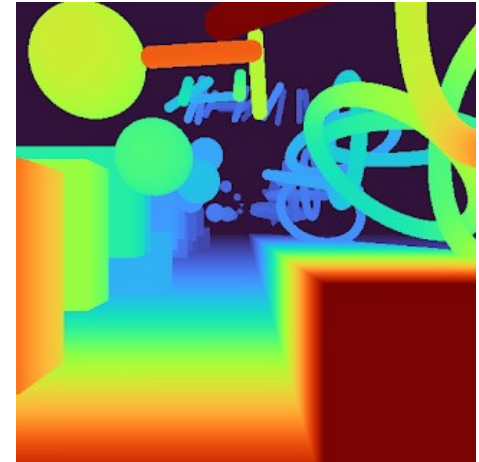
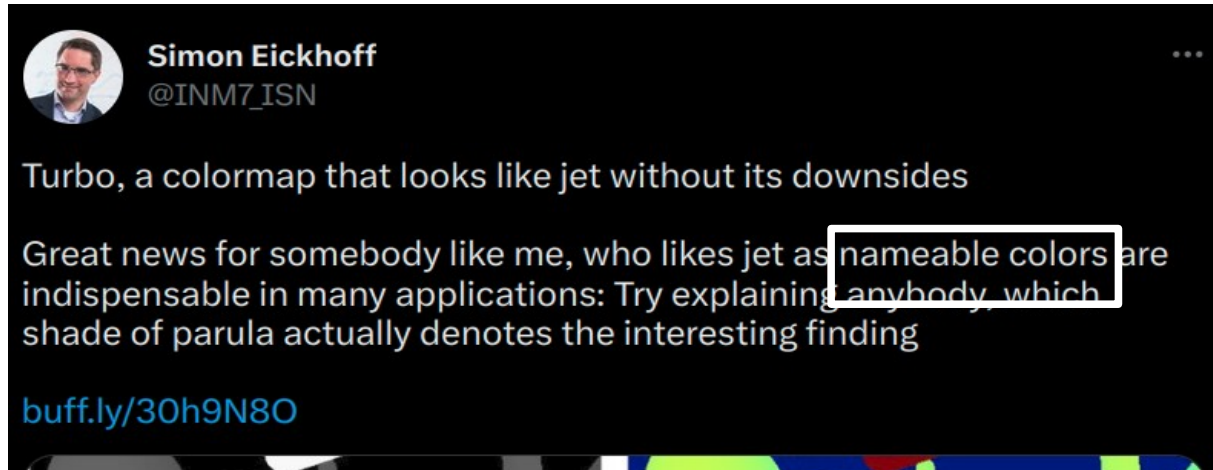
Science: Rainbow colormaps are disliked



Science: Rainbow colormaps are disliked

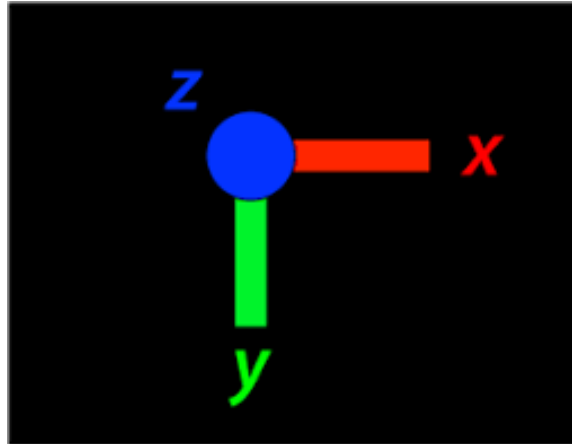
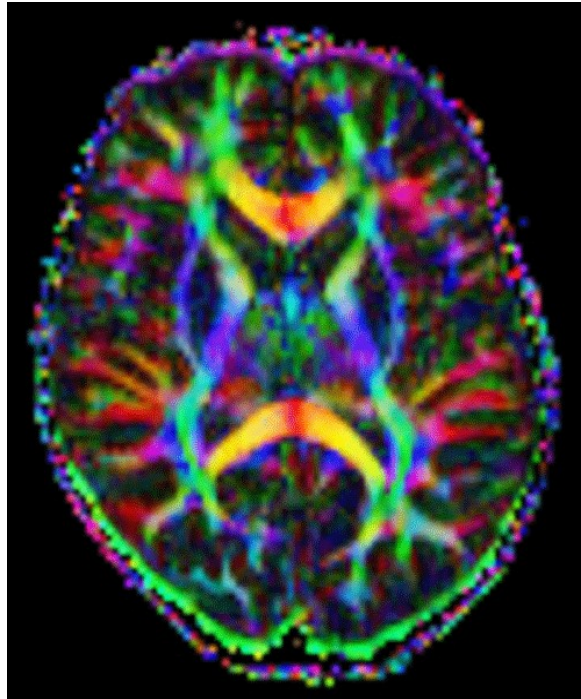


Science: Rainbow colormaps are disliked?



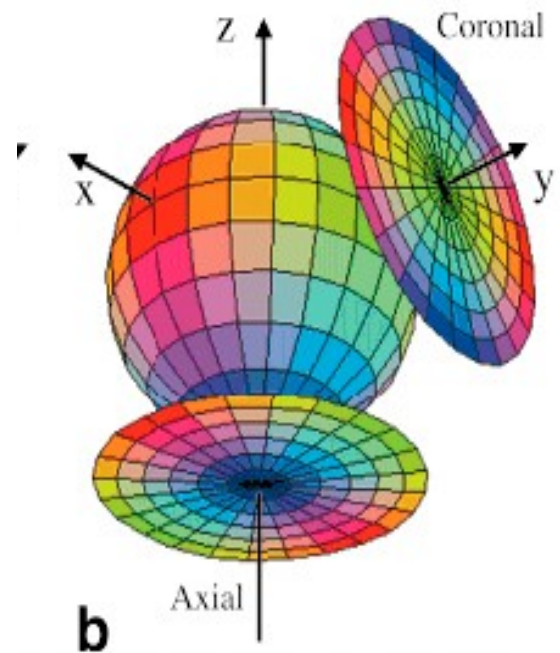
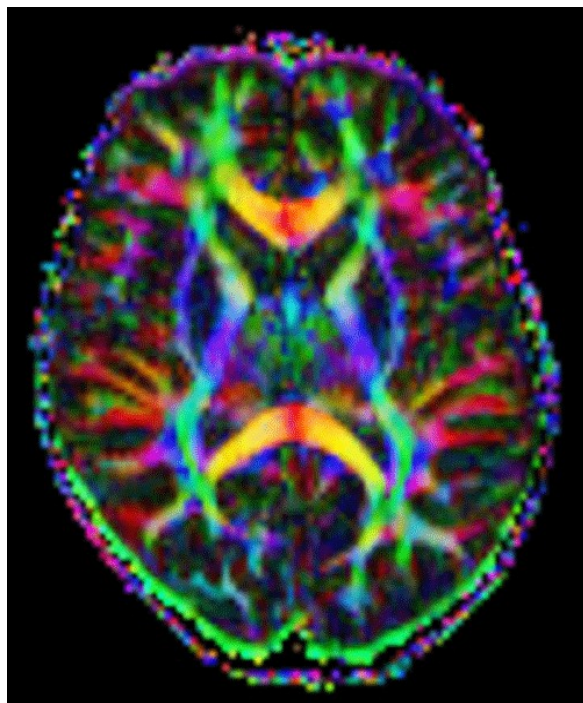
Neuroscience: Colors = 3D direction

Principal Diffusion Direction



Neuroscience: Colors = 3D direction

Principal Diffusion Direction



(For future reference)

Examples of visualization

conventions

- Culture
 - Pink is female
 - Time goes left to right
- Science
 - Rainbow colormaps are disliked
- Neuroscience
 - MRI in grayscale
 - DWI colors = directions

Example 1 – Simple data and figure

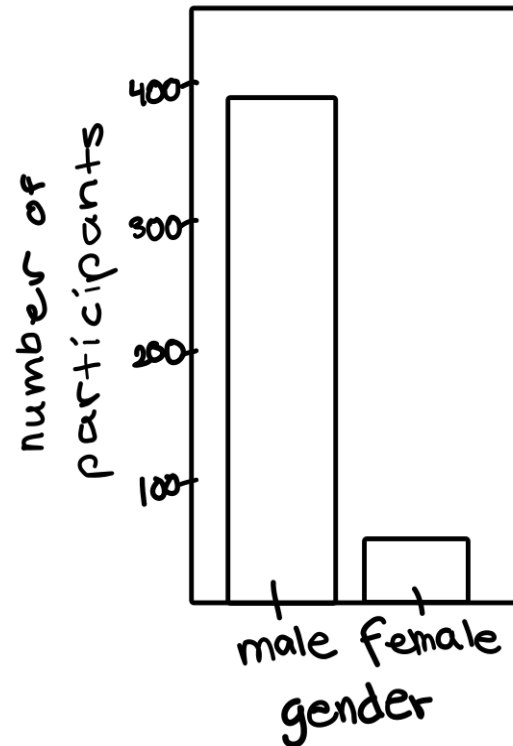
- Original paper on the ABIDE dataset
 - 964 subjects
 - 396 male
 - 51 female
- ← **Lets visualize this**

What's our message?

- More male than female participants
 - 2 categories, 2 values

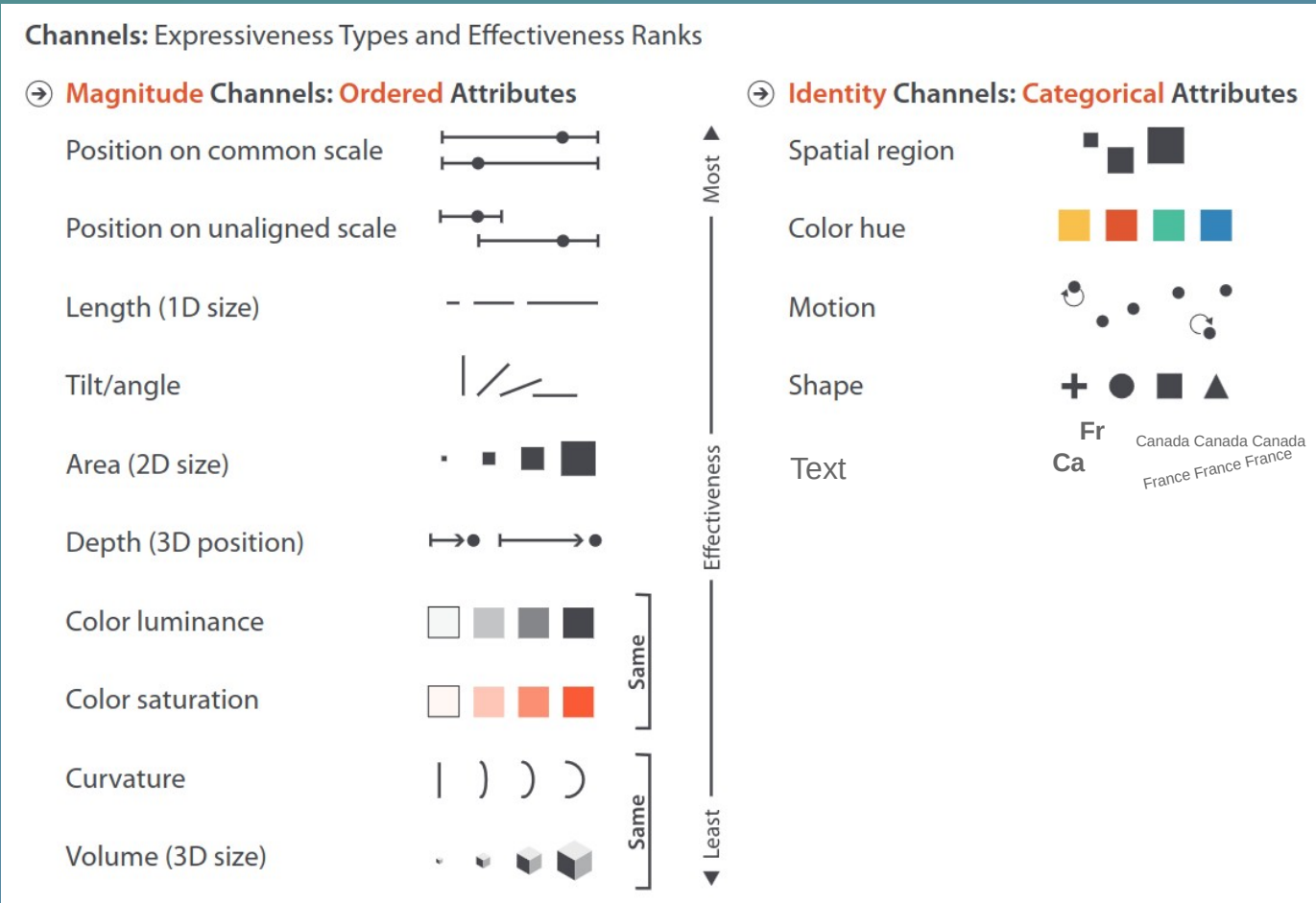
How should we communicate this message?

- Clearly show 2 categories and 2 different values



(For future reference)

Choosing effective charts



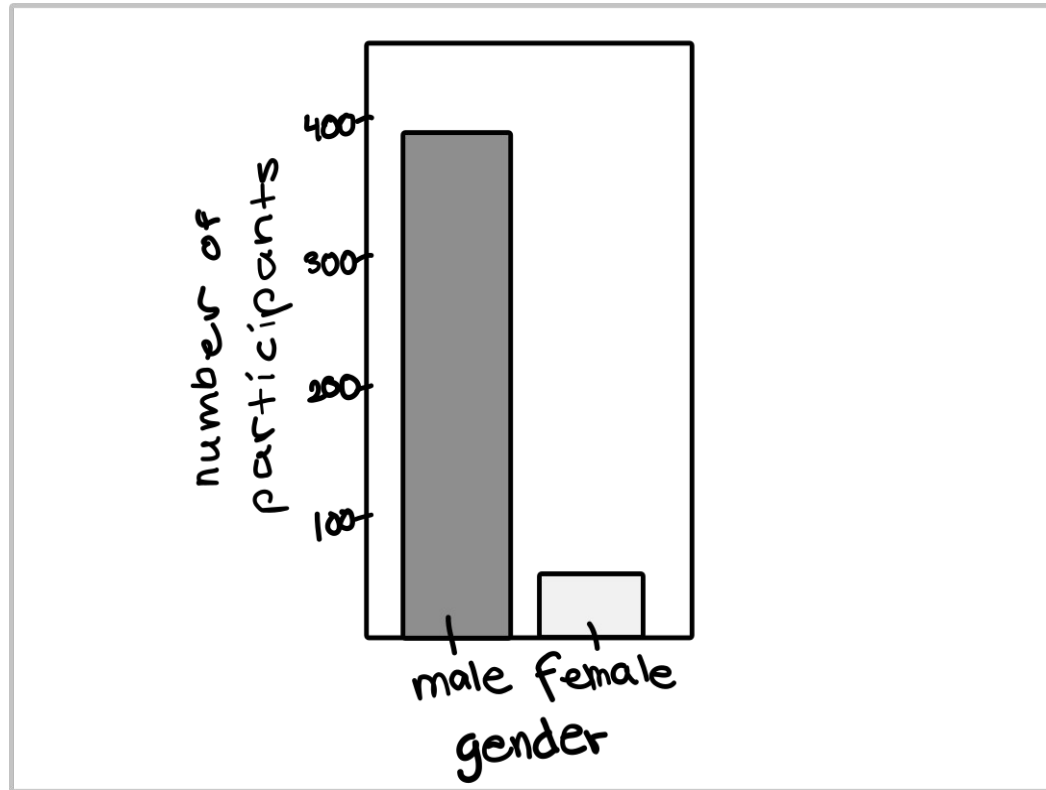
How should we communicate this message?

- Clearly show 2 categories and 2 different values



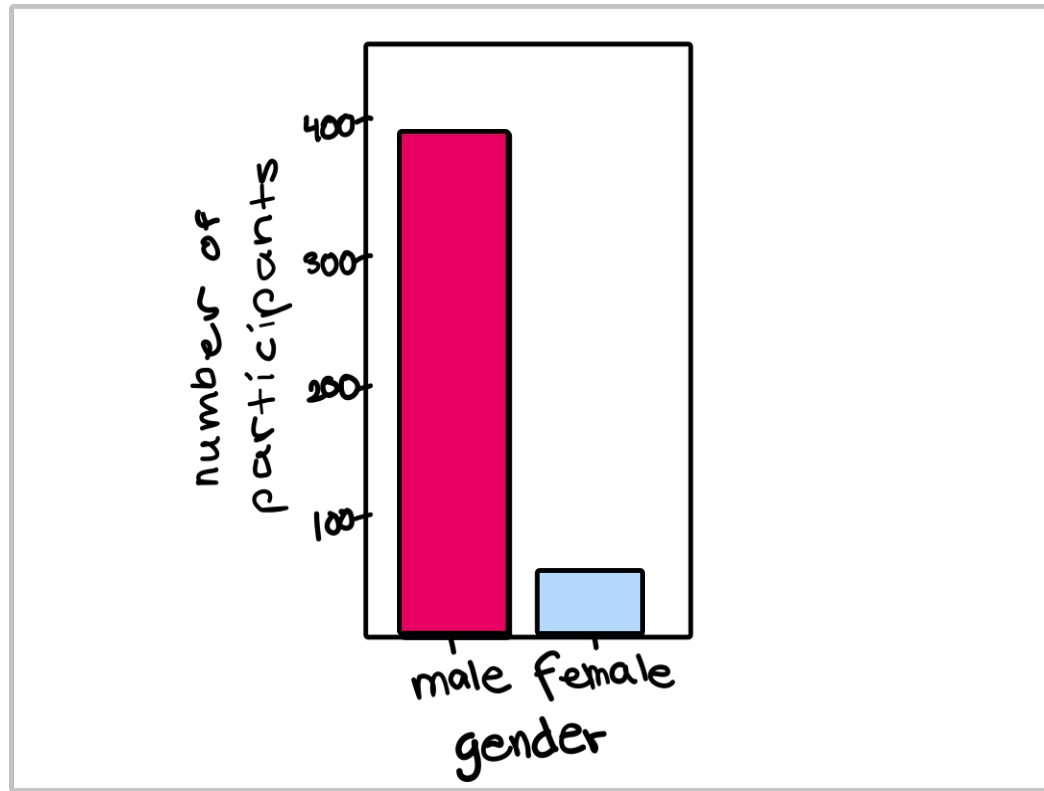
How should we communicate this message?

- Clearly show 2 categories and 2 different **values**



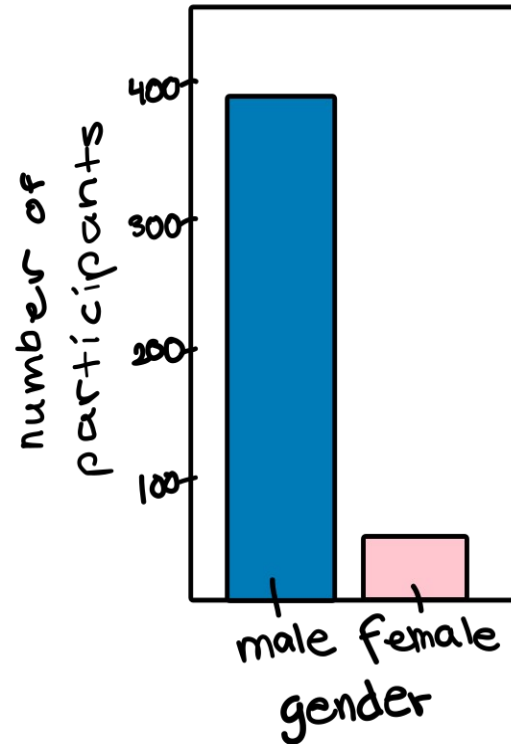
How should we communicate this message?

- Clearly show 2 **categories** and 2 different values



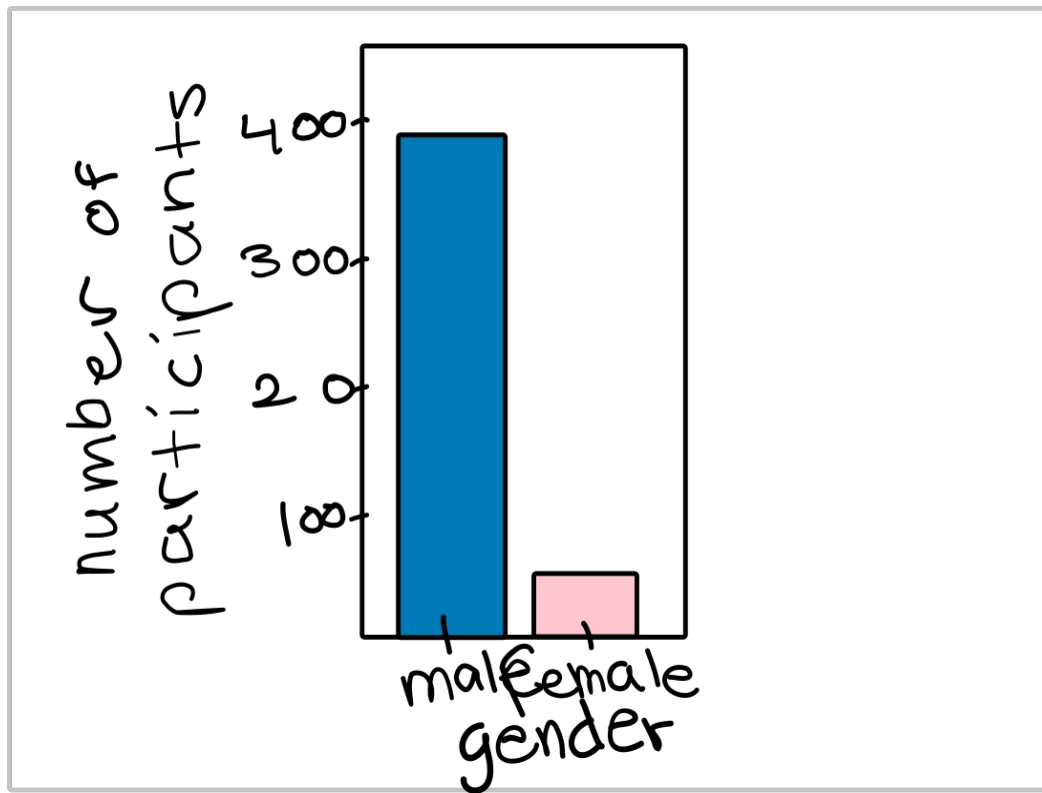
Are there any conventions we should think about?

- Gender & color



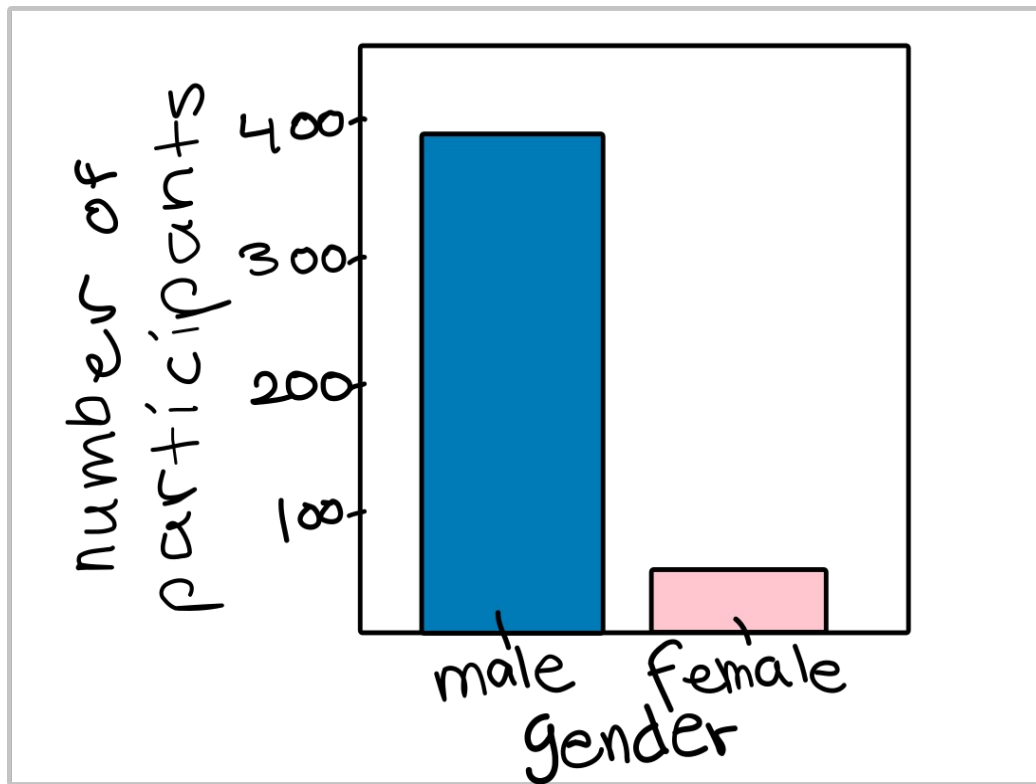
Do we need to adapt it to the context?

- Academic presentation



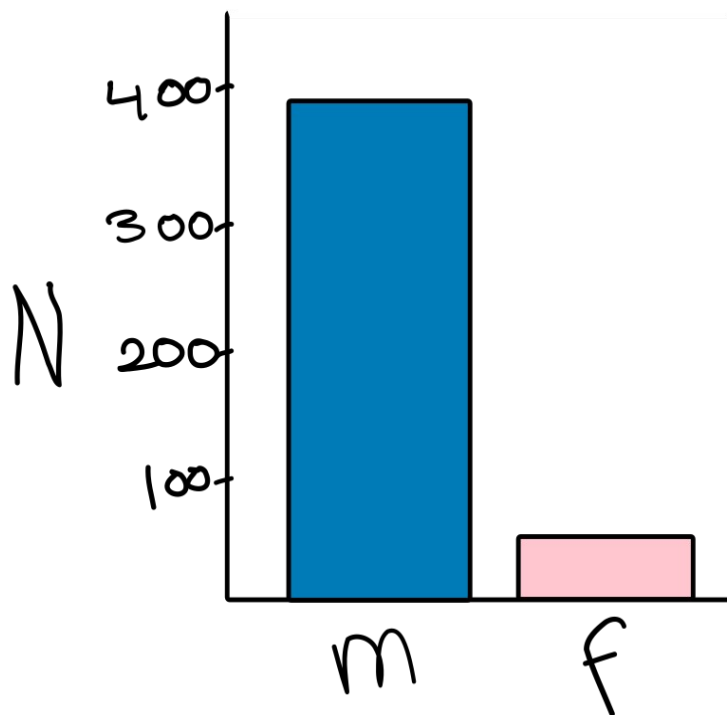
Do we need to adapt it to the context?

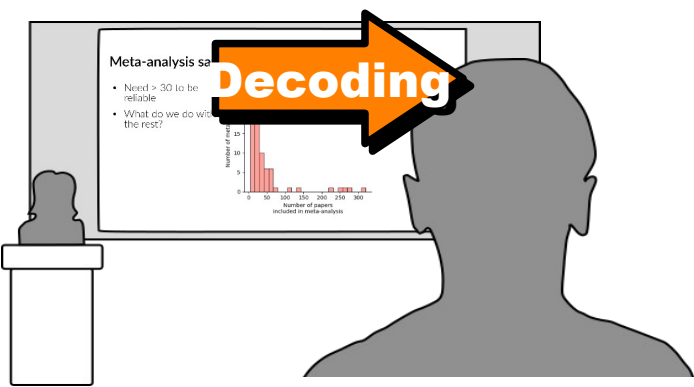
- Academic presentation



Do we need to adapt it to the context?

- Academic presentation





To plan an **effective visualization**, we need to think about

- **Message**

- **What** we want to communicate

- **Perception**

- **How best** to communicate it

- **Conventions**

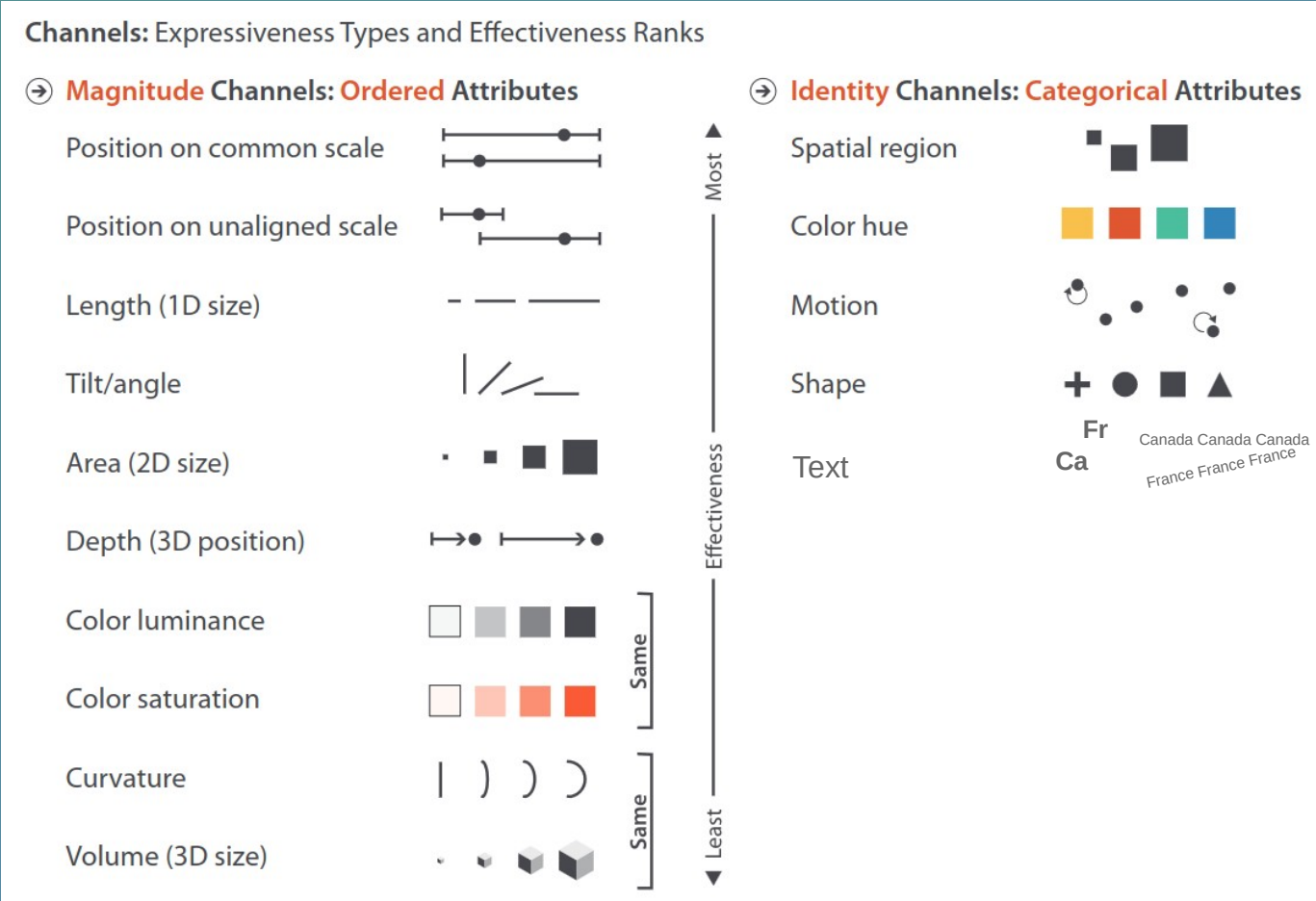
- **How** it's **usually** communicated

The end of part 1

Reference slides...

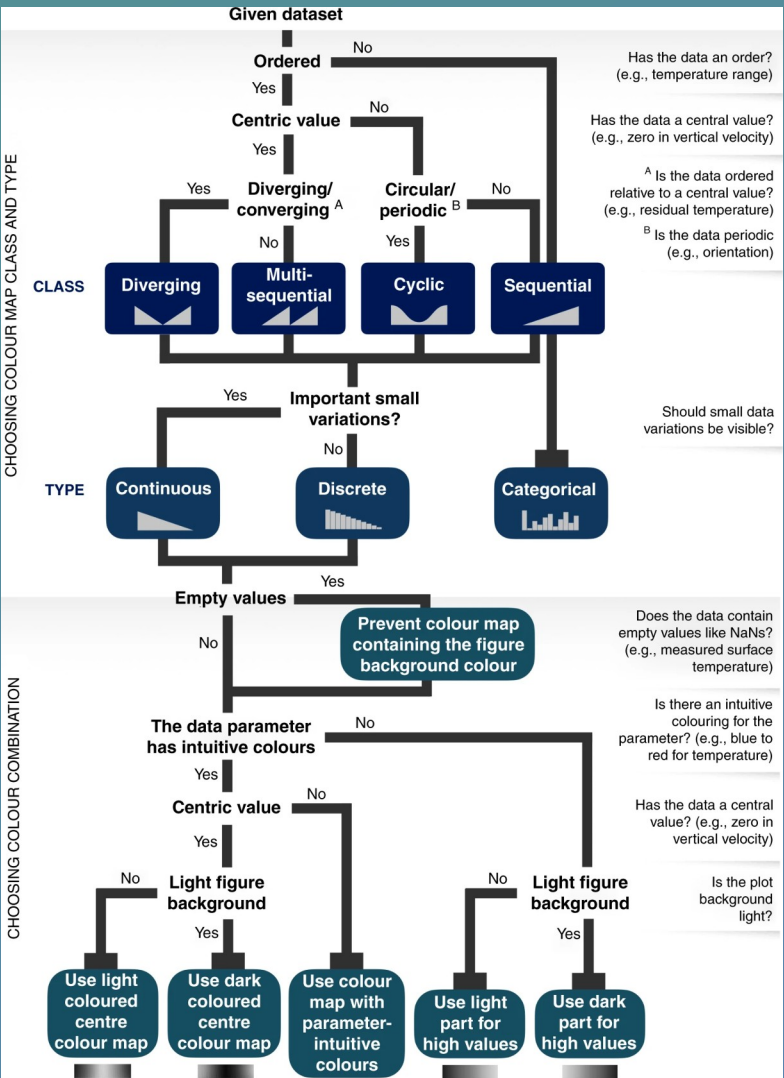
(For future reference)

Choosing effective charts



(For future reference)

Choosing effective colormaps



(For future reference)

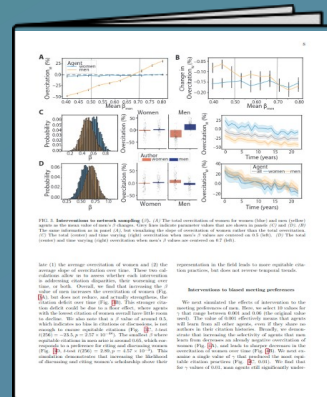
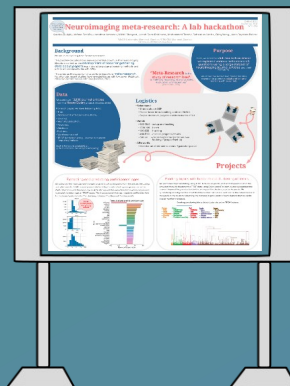
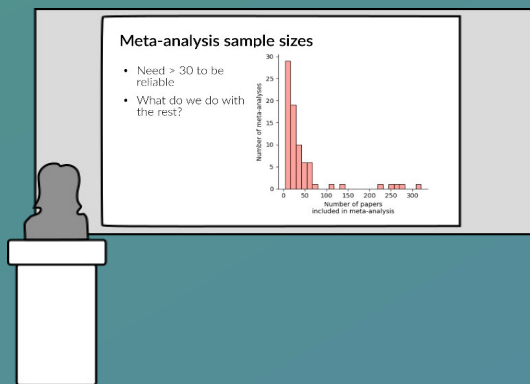
Examples of visualization

conventions

- Culture
 - Pink is female
 - Time goes left to right
- Science
 - Rainbow colormaps are disliked
- Neuroscience
 - MRI in grayscale
 - DWI colors = directions

(For future reference)

Contextual adjustments



Time to understand

Proximity to viz

Text explanations

Self-explanatory

Visual complexity

Less

More